

Expert Platform Recommendations

Architectural improvements, security hardening, code discipline, UI/UX optimizations & high-value missing features.

Focus Areas: Architecture, Code Quality, Security, Backend, UI/UX, Features

Engineering Standard: Ponytail Zero-Waste Discipline

🚨 1. URGENT SECURITY & CORE DATA INTEGRITY

Enforce Row-Level Security (RLS) Policies on Supabase

P0 • SECURITY VULNERABILITY

Currently, all portals rely on client-side filtering with the public anon key. Any student can open DevTools and query `quiz_scores`, `messages`, or `profiles` across the entire school database.

Action Required: Enable RLS on all PostgreSQL tables and enforce `auth.uid()` scoping rules on student scores and conversation messages.

Fix Session Hijacking Vulnerability in Student Chat

P0 • XSS / AUTH BYPASSING

`Chats.jsx` parses `?user=` directly from URL query string and writes it to `localStorage.edtech_user`. A malicious actor can craft a link to hijack any student's role or session.

Action Required: Remove URL parameter user parsing. Authenticate strictly via Supabase Auth token session validation.

Connect Student Academic Progress to Real Backend Data

P0 • DATA INTEGRITY

`Progress.jsx` renders static hardcoded arrays (85%, 68%, etc.), hardcoded 10-day streaks, and permanently locked achievement badges. Scores are also saved to local browser storage only.

Action Required: Create `test_submissions` table in Supabase. Store scores on submit and calculate real trajectory charts and badge triggers.

🏗️ 2. ARCHITECTURAL RESTRUCTURING

Shared Component Monorepo Package

P1 • ARCHITECTURE

Component duplicates (`supabase.js`, `Card.jsx`, `ChatInterface.jsx`) are copy-pasted across 4 separate Vite portals. Create a `packages/shared/` directory with shared hooks and UI primitives.

Convert Study Island 3D to Modular React App

P1 • ARCHITECTURE

`script.js` is a single 2,272-line monolithic JS file. Convert Study Island to a React Vite app where each 3D chapter is a lazy-loaded route component under `/chapters/`.

🌟 3. HIGH-VALUE MISSING FEATURES

1. Student Attendance Tracking System

CORE PLATFORM GAP

Teachers currently have no mechanism to mark present/absent statuses. Build a daily attendance grid for teachers with automatic monthly percentage reports for Admin and Student views.

2. Study Island Gamification & XP System

STUDENT ENGAGEMENT ENGINE

Introduce an XP reward system for completing 3D chapters (+50 XP), 90%+ quiz scores (+100 XP), and 7-day study streaks. Render class leaderboards in Student & Teacher dashboards.

3. AI-Powered Chapter Quiz Generation

TEACHER PRODUCTIVITY

Integrate Gemini API to scan uploaded PDF/text lesson notes and auto-generate 10 multiple-choice questions into the Question Bank with one-click teacher approval.

4. Web Push Notification Engine

NATIVE BROWSER API

Utilize the native browser Web Push API to send instant alerts when a live class starts, a new test is assigned, or a direct message is received—even when the tab is closed.

📋 SUMMARY PRIORITY ACTION MATRIX

PRIORITY	CATEGORY	ITEM / FOCUS AREA	EXPECTED OUTCOME
P0 Urgent	Security	Supabase Row-Level Security (RLS)	Blocks unauthorized database reads/writes in DevTools
P0 Urgent	Security	Remove URL ?user= parsing	Eliminates stored XSS and role spoofing vulnerabilities
P0 Urgent	Data Integrity	Real Progress engine & Test submissions	Stores quiz attempts in DB & renders real score graphs
P1 High	Architecture	Shared packages/shared/ Monorepo	Single source of truth for Supabase client & shared UI
P1 High	UI / UX	Mobile Responsive Layouts	Enables mobile usage for students across India
P1 High	Feature	Student Attendance System	Completes essential school administrative suite
P2 Medium	Feature	Study Island Gamification & XP	Drives daily student retention and chapter completion
P2 Medium	Feature	AI Quiz Generation (Gemini API)	Automates question generation for teachers